Alfat Tasnim Hasan

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Professional Summary

Researcher, web developer, and trainer with experience in Machine Learning, Deep Learning, and full-stack software development. Worked on MERN and Flask-based applications, as well as AI-driven predictive systems using explainable AI techniques such as SHAP and LIME. Research papers have been published and presented at IEEE conferences in areas such as mental health prediction and personality analysis. Skilled in Python, React, TensorFlow, and data analytics, with a strong interest in developing practical and user-focused technology solutions through research and innovation.

Technical Skills

- **Programming Languages:** C, C++, Python, Java, JavaScript, PHP (Basic)
- **Web Development:** HTML, CSS, Tailwind (Basic), React, Node.js, Express, Flask
- **Machine Learning:** Scikit-learn, TensorFlow, Supervised & Unsupervised Learning
- **Deep Learning:** Neural Networks, CNN, RNN, Model Training, Hyperparameter Tuning
- **Data Tools:** Pandas, NumPy, Seaborn, Matplotlib, Data Preprocessing, Feature Engineering
- **Model Explainability:** SHAP, LIME
- **Databases:** MongoDB, MySQL
- **Tools:** Git, VS Code, Code::Blocks, IntelliJ IDEA Community, PyCharm, Postman, Jupyter Notebook, Anaconda, XAMPP, Linux (Ubuntu), MongoDB Atlas
- **Office Tools:** Microsoft Office, Odoo
- **Core Concepts:** Data Structures, Algorithms, Statistics, Model Optimization, Problem Solving

Core Skills

- Teamwork and Collaboration
- Educational Technology and Digital Learning Tools
- Time Management and Organizational Skills
- Adaptability and Quick Learning
- Problem-Solving and Critical Thinking
- Professional Ethics and Positive Attitude
- Lesson Planning and Curriculum Implementation

Professional Experience

Software Developer Apr 2026 – Present
Nexogs Systems Ltd., *Dhaka, Bangladesh*

- Developed and deployed full-stack MERN applications, improving system performance and user experience.
- Optimized application performance, debugged complex issues, and implemented scalable software solutions.
- Designed and maintained databases while integrating third-party and internal APIs to ensure reliable system functionality.

Software Developer Intern Dec 2025 – Mar 2026
Nexogs Systems Ltd., *Dhaka, Bangladesh*

- Assisted in the development and deployment of full-stack MERN applications.
- Collaborated with senior developers to optimize application performance and resolve technical issues.
- Participated in database design, API integration, and software testing to ensure application reliability.

Part-time Research Assistant Jan 2025 – Present
Under the supervision of a faculty member, **Anika Rahman**, **Stamford University Bangladesh**, Dhaka, Bangladesh

- Assisted a faculty member in preparing journal and conference publications using ML/DL models.
- Developed prototypes and interactive web demos using Python and Flask.
- Performed data preprocessing, model training, and evaluation to support research projects.
- Collaborated closely with the faculty member to implement solutions and ensure timely publication.

Projects

Nexogs Systems Ltd. – Official Company Website

- Built and contributed to full-stack MERN-based web applications with scalable architecture.
- Implemented authentication, API integration, and performance optimization features.
- Collaborated with the development team to design and deploy production-level software solutions.
- Live Deployment: https://nexogs.com/

BLAST Legal Helpdesk– Official CRM module

- Developed and contributed to a full-stack MERN-based CRM application with a scalable and maintainable architecture.
- Implemented authentication, API integration, and performance optimization features.
- Collaborated with the development team to design, develop, and deploy production-ready software solutions.
- Live Deployment: https://helpdesk.blast.org.bd/

NutriAI – Nutrition Prediction and Management

- Full-stack web app using Python Flask, HTML, CSS, and JavaScript.
- ML-based system for personalized nutrition prediction and recommendations.
- Live Deployment: https://nutriai-xco6.onrender.com/

Jahanabad Military Driving School – Educational Portal

- MERN-based web portal with interactive features and user-friendly design.
- Live Deployment: https://jmdschool.com/

Devil's Den – E-commerce Frontend

- Frontend-only MERN project showcasing public pages and UI design.
- Deployed: https://devil12.netlify.app/

Publications

Descriptor: A Dataset on Educational, Lifestyle, and Socioeconomic Factors of Bangladeshi Students (DELSFBS)
IEEE Journal Article
DOI: 10.1109/IEEEDATA.2026.3658067
This paper presents a curated dataset of Bangladeshi students, capturing educational, lifestyle, and socioeconomic attributes. The dataset facilitates the analysis of relationships between academic performance and behavioral factors, enabling data-driven research and machine learning applications in education and social sciences.

Mental Health Risk Assessment in University Students: Integrating IoT Environmental Data with Hybrid Machine Learning
IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS 2025)
DOI: 10.1109/COMPAS67506.2025.11381719
This study proposes a hybrid machine learning framework for predicting mental health risk levels among university students using IoT-based environmental data. An ensemble model combining Random Forest and Support Vector Machine (RBF kernel) achieved 98% classification accuracy across Low, Medium, and High risk categories. The findings demonstrate the effectiveness of real-time environmental sensing for early mental health risk detection and smart campus applications.

From Data to Deployment: A Personality Prediction Pipeline with SHAP-Based Feature Insights and Flask Implementation
IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2026)
DOI: 10.1109/QPAIN69676.2026.11546412
This paper presents an end-to-end personality prediction pipeline, integrating data preprocessing, machine learning modeling, and deployment using Flask. It incorporates SHAP-based interpretability techniques to analyze feature contributions, enhancing model transparency and enabling practical, real-world deployment of explainable AI systems.

Education

B.Sc. in Computer Science and Engineering 2022 – 2025
Stamford University Bangladesh, Dhaka
CGPA: 3.78 / 4.00

Higher Secondary Certificate (HSC), Science 2017 – 2019
Dawood Public School and College, Jashore
GPA: 4.17 / 5.00

Secondary School Certificate (SSC), Science 2007 – 2017
Jashore Zilla School, Jashore
GPA: 4.95 / 5.00

Certifications & Achievements

Python (Basic) – HackerRank
Completed a beginner-level Python programming course focusing on fundamentals and problem-solving.

Certificate of Reviewing – Mental Health & Prevention Journal
Certificate awarded in recognition of reviewing and contributing to the peer-review process for a manuscript submitted to the Mental Health & Prevention journal.

Mental Health Risk Assessment in University Students – IEEE COMPAS 2025
Certificate for conference paper publication and presentation on integrating IoT data with hybrid ML models.

From Data to Deployment: A Personality Prediction Pipeline with SHAP-Based Feature Insights and Flask Implementation – IEEE QPAIN 2026
Certificate for conference paper presentation on explainable AI-based personality prediction using SHAP interpretability and Flask deployment.

Inter-Campus Programming Contest 2025 – Stamford University Bangladesh
Received a Team Certificate in an Intra-Campus Programming Contest.

Admission Fair 2022 Participation – Stamford University Bangladesh
Certificate for active participation in extracurricular activities during admission fair.

IEEE CS BDC Summer Symposium 2025 – Hajee Mohammad Danesh Science and Technology University (HSTU)
Certificate for presenting own research at a student-led IEEE symposium.

16th National Undergraduate Mathematics Olympiad 2025 - East West University
Participated in a Nationwide Undergraduate-Level Mathematics Competition.

Machine Learning for Beginners – Sololearn
Gained foundational knowledge in machine learning concepts and basic model implementation.

Java Workshop for Absolute Beginners – Hands-on Workshop
Introduced to Java programming fundamentals with practical exercises.

Build Your First Project with PHP – Ostad
Learned to develop a small PHP project, covering core PHP concepts and application logic.